Engineering Chemistry · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry

DRAFT — UE26CY151A · Unit_01 · Molecular spectroscopy, phase equilibria, and computational chemistry · Interaction of radiation with matter

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
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Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Interaction of radiation with matter

1 Interaction of radiation with matter

Study focus: Interaction of radiation with matter. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000002 · EV0000003

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Dr.

Key details

- Lata Pasukuleti and I'll be teaching you this course on engineering chemistry.
- The content is like this.
- First we are going to look at interaction of electromagnetic radiation with matter, then electromagnetic spectrum, then we discuss the Born Oppenheimer approximation.
- Further we study BL Ambert's law and then we look at different types of spectroscopy.
- First we'll study microwave spectroscopy, then we go to IR spectroscopy and finally to electronic spectroscopy.
- In today's class, we will look at the definition of molecular spectroscopy, what we understand by interaction of electromagnetic radiation with matter, comparison between atomic and molecular spectrum.
- The quantization of energy concept, absorption and emission spectrum, and finally about spectroscopic units.
- The word spectroscopy has the in it spectrum.
- to study molecules.
- This happens because whenever we have a sample, say like this, a sample.
- And then we have electromagnetic radiation falling on this sample.
- The electromagnetic radiation has energy and this radiation is going to interact with the matter here.
- Now because of this interaction, there are bound to be changes in the electromagnetic radiation.

- Some of the energy could be absorbed, some of it could be transmitted, and some of it could get scattered.
- Nature of the matter.
- Here we are talking of molecular spectroscopy, so we are going to study how it is affecting the molecules.
- This interaction of electromagnetic radiation with matter is the concept which we are going to use in molecular spectroscopy.
- I need incident radiation.
- These radiation and this radiation is going to fall on the matter here, the molecules here, and there is going to be interaction.
- For that I need a detector or an analyzer we call it.
- Going to send this radiation to a diffraction grading.
- Because it's going to break the radiation into the different components of different wavelengths.
- And then we need a recorder which is going to record these different wavelengths, and what we get is.
- Known as a spectrum.
- So that means if radiation falls on a sample and we are able to analyze the radiation which is coming out, then we get what we call as a spectrum.

Example or application from the lecture

- So first let us define what is molecular spectroscopy.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000002 · EV0000003 · EV0000004 · +1 source units in pack.json

Core explanation · 05:00–09:59

It is a stream of particles which are called.

Key details

- Photons and these are moving through space in the form of waves.
- The radiation is going to have particle, we will see particle nature as well as wave nature in the form of we can call it as a stream of particles.
- particles called photons or we can look at it as a form of a wave.
- Radiations and they are oscillating perpendicular to each other and also perpendicular to the direction of propagation.
- electric field oscillating and we also have the magnetic field which is perpendicular and these are going to be perpendicular to the direction of propagation we can see in the picture over here and all electromagnetic radiations always travel with the velocity of light.
- These are important features which we should remember when we are studying molecular spectroscopy.
- C, what are the characteristics of the wave or how is a wave characterized?
- What are the various properties that a wave possesses?

- distance between two successive crests and troughs of the wave and it is it is denoted by lambda and it could be in meters, nanometers or angstrom depending on what is the distance between two successive casts or traps.
- Then we also have frequency.
- You can make out from the unit itself.
- minus thirty four Joues second and nu is the frequency.
- And wavelength is related to frequency by C is equal to nu lambda, c is the speed of light, lambda is the wavelength, and nu is the frequency.
- And there is also wave number which is related to.
- Wavelength by nu bar is equal to 1 by lambda.
- That's exactly what I said.
- All these properties, wave properties we should understand.
- Atomic spectrum is something which you have already studied.
- See, we are going to learn about molecular spectrum, but if you recollect what you studied in atomic spectrum, then understanding molecular spectr would be that much more simple.
- And these are the atomic orbitals which are present around the atom.
- This has just one electron in it, and we know that when radiation is going to fall on.
- This hydrogen atom, then these elect this electron is going to be excited to different levels.
- Which level it goes to depends on what is the amount of energy it absorbs.
- From one to three, one to four.

Example or application from the lecture

- What is frequency the number of cycles completed in a certain amount of time and that is given by nu, and it could be in hertz or the second inverse.
- We also have wave number for every wave and that is number of complete waves or cycles that is contained in a unit distance and that is given by new bar and that is centimeter inverse.

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Core explanation · 10:00–14:59

In to any of these higher energy orbitals, the electron transits when it absorbs radiation.

Key details

- In this case, it is atom.
- If we can record, we will get a spectrum like this.
- One, two, three, third level.
- Some will come to the first level.
- Some will come back to the second level.
- And the difference in energy for each of this.
- When it comes from here to here energy is different.
- If energy is different, the frequency and the corresponding wavelength are also different.
- Based on this difference in energy, we will get different lines in the spectrum.

- Electron coming back from the higher energy level to the lower energy level.
- One thing we can understand here that if we see a line in the spectrum.
- This is something you're physically observing.
- So if we see a line in the spectrum, it means that corresponding to that frequency, there is an energy gap which is present in the.
- In the atom, and that energy gap is between atomic orbitals because we are talking of atoms, it is between the atomic orbitals.
- The gap is equivalent to the energy which is getting emitted.
- In this case, this is an emission spectrum.
- So if you see a line in the spectrum, it means that there exist that two energy levels of that energy gap.
- That is about atomic spectrum and this spectrum is always is known as the line spectrum because you will see different lines corresponding to the energy emission which is happening here.
- Just like you have atomic orbitals, you have studied molecular orbitals also.
- Molecular orbitals exist inside the molecule just like in an atom there are atomic orbitals.
- That is known as the Homo, highest occupied molecular orbital.
- The electron now is going to transit from the highest or occupied molecular orbital to lowest unoccupied molecular orbital.
- And when such a transition happens, then we will see a line in the spectrum.
- That is because.
- Energy gap is there, that much energy is absorbed, say we are looking at absorption, then we'll find that particular line in the energy is the line corresponding to that particular energy is missing in the spectrum.
- The molecule above.
- Absorb that energy and transit it to the next level, and that is known as the lowest unoccupied molecular orbital.
- And this transition is to the lowest unoccupied molecular orbital.
- This is what we are going to see in our molecule.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000010 · EV0000011 · EV0000012 · +1 source units in pack.json

Core explanation · 15:00–19:59

We should see a line in the spectrum, right?

Key details

- But we find that in the case of molecules, we don't see a line.
- In fact, we see a band like this.
- If you look closely, if you have good resolution in your spectral photometer, you will find that these this band is actually made of several lines inside.
- What is this band spectrum in a molecule?
- We understand that in a molecule apart from electronic transition there could be other types of transitions possible.

- Because a molecule can undergo vibration.
- It can also undergo rotational rotation.
- So, therefore, apart from the electronic transition, there could be other transitions to different levels, like there could be change in vibrational levels, there could be change in rotation levels.
- An atom you have only atomic orbital, so you will see only electronic transition.
- There's only one line which you will see you will see only lines in the atomic spectrum, but in a molecule, you are going to see a band, and if you have good resolution, you will see finer lines inside this band.
- As I mentioned, if you see a line in the spectrum, that means there are corresponding energy levels which are equal to the energy gap.
- The frequency of the absorption or the emission which you see in the spectrum.
- Therefore, what do we understand so far?
- That atomic spectrum will give you a line spectrum, whereas molecular spectrum will give you a.
- molecular spectrum will be known as a band spectrum.
- You will see a band with finer lines.
- We know that energy levels in an atom or a molecule are not continuous, instead they are discrete or quantized.
- What do peep what do we mean by that?
- that we have energy levels at certain values only.
- It's not a continuous energy that is available inside the atom or a molecule.
- Like in a ladder you have rungs, different rungs.
- I you can put your hand leg on one rung or the next rung and the next rung.
- You cannot stand if you put your leg in between these rung.
- Exactly like that in an atom, you have these orbitals energy levels in a molecule also energy levels which are at discrete energy value.
- And then such an energy level is known as a quantized energy level.
- It is not continuous, it is quantized.
- Now because it is quantized, that means it will there will be a level here, there will be a level here.
- There cannot be any levels in between.
- Therefore, can be absorbed only in the form of packets which are known as quanta.
- By H nu.
- H is the Planck's constant and mu is the frequency of the radiation which will get absorbed or emitted.
- So that means energy is quantized means this h nu can be.
- Either as only H new or it could be two H new, three H new, four H new, but you cannot have a quarter H nu or a th three by fourth H new.
- No, that is what we mean by energy is quantities means the energy gap is always equal to some integral multiple.
- Of H nu.
- to this energy level and this difference in radiation is H nu.

Example or application from the lecture

- You can see this is another example where you're actually seeing a band.
- This energy gap is will be suppose there is an energy level E1 and there's an energy level E2, the energy gap delta E is equal to E2 minus E1.
- This difference in energy will be what we say quantize and therefore that is given.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000014 · EV0000015 · EV0000016 · +1 source units in pack.json

Core explanation · 20:00–24:59

Corresponding to the energy gap, the frequency of light is going to be absorbed.

Key details

- One is absorption spectrum, the other one is emission spectrum.
- What would be an absorption spectrum when light is absorbed.
- This would give you an absorption spectrum.
- The lower energy level to higher energy level, thereby absorbing a particular radiation.
- Absorbing a particular radiation would be like you will see some lines if you have used white light, then you would see some lines missing.
- That means there is a corresponding energy, two energy levels with that corresponding energy gap.
- That is known as an absorption spectrum, you can see here.
- And if the light has already been absorbed and now it is it's coming the atom or molecule is coming from the higher energy level to the lower energy level then what we see is known as the emission spectrum.
- So basically what it means that you can up it depends on the detector whether you're studying the absorption spectrum or you're studying the emission spectrum.
- Again, it tells us that based on the energy gap, light is going to be absorbed or emitted.
- It depends on what we are what is the detector, how we are analyzing or what spectrum we want to study.
- And most of our study here, we will be looking at absorption spectrum.
- Is expressed in joules.
- We know this relation also.
- Now, since c is equal to nu lambda, so nu bar can be expressed as nu by c, and therefore energy can be expressed in terms of nu bar.
- as HC nu bar.
- Why do we do that?
- Because new bar is treated as the spectroscopic unit of energy and basically it just tells us it corresponds to energy of a radiation and it is in terms of centimeter inverse.
- If we express why should we express in centimeter inverse and not in joules?
- It is just a matter of convenience.
- Because if you see you can try to do this conversion, one centimeter inverse is equal to 1.99 into 10 to the power of 23 joules.
- Minus 23 joules.
- So therefore 1 centimeter inverse is 1.99 tet power minus 23 joules.

- Amount of energy.
- Basically, we don't want to use such exponential terms, large exponential terms.
- Therefore, we can express in a simple way, we just say one centimeter inverse, five centimeter inverse, it's just a matter of convenience.
- What it means is this energy in joules.
- Has been expressed in terms of centimeter inverse.
- Though we know centimeter inverse is not an SI unit, but still it is for convenience that we will express energy in terms of centimeter inverse, and therefore it is referred to as the spectroscopic unit for energy.
- And before that, let me tell you that in the next class we are going to study about electromagnetic spectrum.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000018 · EV0000019 · EV0000020 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Interaction of radiation with matter transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Interaction of radiation with matter transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Interaction of radiation with matter transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Interaction of radiation with matter transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_01/AV_Summaries/001_Interaction_of_radiation_with_matter__1eb05f6c3f230eec.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00002	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00003	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00004	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...
SRC00005	5 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0042. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0043 to s0062. Sidecar variants compare...

C Revision and retrieval practice